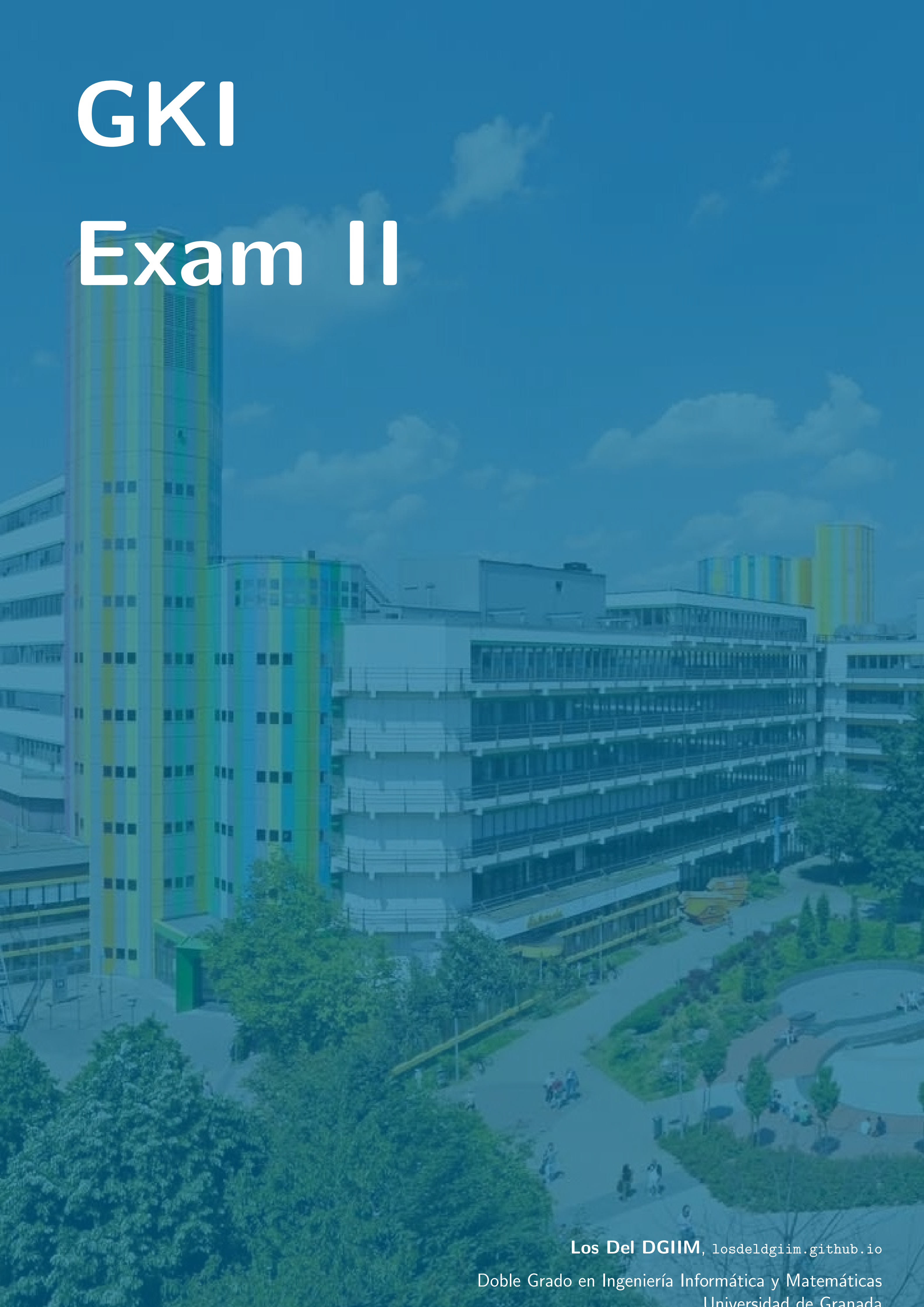

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Exam II

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Doble Grado en Ingeniería Informática y Matemáticas

Universidad de Granada

GKI
Exam II

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Arturo Olivares Martos

Granada, 2026

Subject Fundamentals of Artificial Intelligence.

Academic Year Summer Semester 2023.

Lecturer Gérald Kämmerer.

Description Final Exam.

Date 01.08.2023.

Duration 90 minutes.

Part 1: Short Questions (54 pts.)

Ejercicio 1 (Unsupervised Learning (3 pts.)). What is unsupervised learning? Mark whether each of the following methods belongs to unsupervised learning or not.

Method	Right	Wrong
PCA	☐	☐
Hill Climbing	☐	☐
kNN	☐	☐
kMeans	☐	☐
t-SNE	☐	☐
GANs	☐	☐

Tabla 1: Classification of Unsupervised Learning Methods

Ejercicio 2 (KDD Process (2 pts.)). What are the steps involved in the Knowledge Discovery in Databases (KDD) process?

Ejercicio 3 (Date Conversion Issues (1 pt.)). Indicate a possible problem when converting the date 08/01/23.

Ejercicio 4 (Basic Statistics (3 pts.)). Given the numbers $A = \{3, 4, 5, 10, 11, 1, 8\}$, compute the median, mean, and variance.

Ejercicio 5 (Data Aggregation Problems (2 pts.)). What problems can occur when performing data aggregation? Mark whether the statement is right or wrong.

Problem	Right	Wrong
Information gain	☐	☐
Information loss	☐	☐
Data reduction	☐	☐
Data separation	☐	☐

Tabla 2: Data Aggregation Issues

Ejercicio 6 (Curse of Dimensionality (2 pts.)). Where does the effect of the "Curse of Dimensionality" occur?

Method	Right	Wrong
High-dimensional search	☐	☐
Data aggregation	☐	☐
Optimize a neural network	☐	☐
Execute a binary classifier	☐	☐

Tabla 3: Curse of Dimensionality Contexts

Ejercicio 7 (Search Methods (1 pt.)). For which search method is Depth-Limited Search a modification?

Ejercicio 8 (Topological Search (5 pts.))**.** Give an example to explain Topological Search.

Ejercicio 9 (Genetic Algorithms - Mutation & Crossover (3 pts.))**.** Given two genes of length 10 with bases $\{0, 1, 2\}$:

- Gene 1: 0101211221
- Gene 2: 2111111001

Apply mutation and a crossover type to these genes starting with Gene 1.

Ejercicio 10 (Uniform Cost Search (4 pts.))**.** Explain Uniform Cost Search using the graph in Figure 1. Go through the different steps.

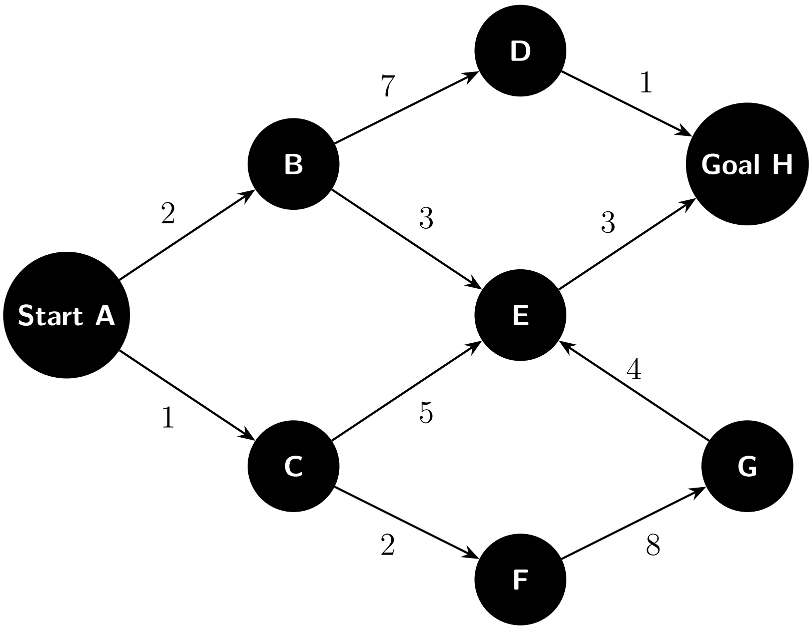

Figura 1: Uniform Cost Graph

Ejercicio 11 (Roulette Selection (2 pts.))**.** What is Roulette Selection in genetic algorithms?

Ejercicio 12 (Underfitting vs. Overfitting (2 pts.))**.** Explain what is meant by underfitting and overfitting in machine learning.

Ejercicio 13 (Confusion Matrix Elements (6 pts.))**.** Fill in the missing entries correctly in Table 4 and explain the elements of the confusion matrix.

Predicted / Actual	**Actual Yes**	**Actual No**	**Actual Total**
Predicted Yes	45 %	15 %	
Predicted No		15 %	40 %
Predicted Total	70 %		

Tabla 4: Confusion Matrix

Ejercicio 14 (F1-Score Definition (1 pt.)). How is the F1-Score defined?

Ejercicio 15 (Resampling Methods (2 pts.)). Name one resampling method and explain how it works.

Ejercicio 16 (Linear vs. Ridge vs. Lasso (3 pts.)). List the differences between Linear Regression, Ridge Regression, and Lasso Regression.

Ejercicio 17 (Principal Component Analysis (1 pt.)). What happens during Principal Component Analysis (PCA)?

Ejercicio 18 (Top-Down Approach (2 pts.)). What is the top-down approach in hierarchical clustering or decision trees?

Ejercicio 19 (Linkage Types in Hierarchical Clustering (4 pts.)). Name two types of linkage in hierarchical clustering and state the mutual advantages and disadvantages of each.

Ejercicio 20 (Activation Function Definition (1 pt.)). Name an activation function and give its mathematical definition.

Ejercicio 21 (Feedback Connections in Neural Networks (3 pts.)). Draw and label a lateral feedback connection, a direct feedback connection, and an indirect feedback connection in the network given in Figure 2.

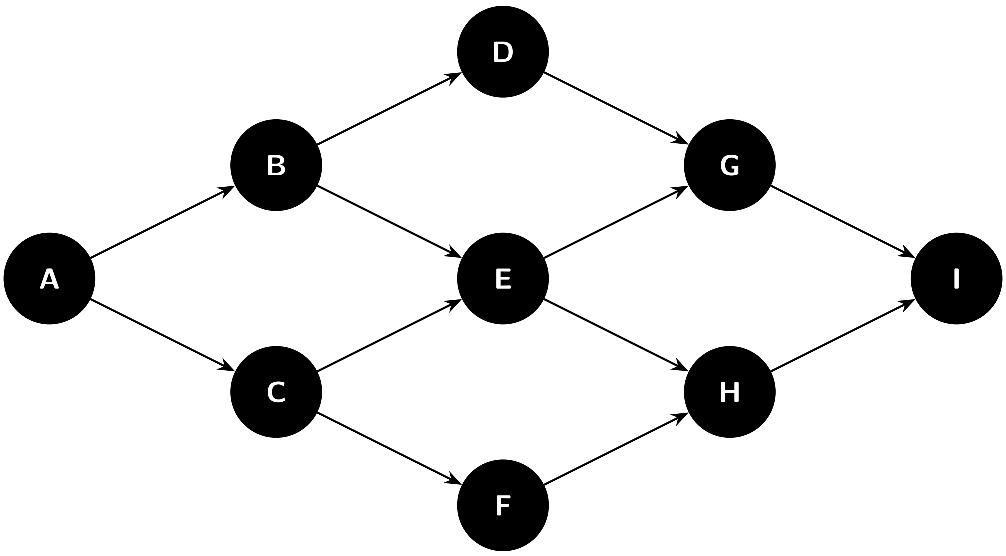

Figura 2: Forward-Feeding Network

Ejercicio 22 (Stemming (1 pt.)). What is stemming in Natural Language Processing (NLP)?

Part 2: kNN Classification (15 pts.)

The precipitation dataset (Table 5) is provided. The goal is to classify the two days from Table 6 as rainy or non-rainy days.

Day	**Max. Temperature**	**Cloudiness**	**Rain Level**	**Rainy Day**
1	20,1°C	30 %	0.30 mm	Yes
2	29,8°C	80 %	0.28 mm	Yes
3	21,2°C	25 %	0.00 mm	No
4	25,0°C	20 %	0.01 mm	No
5	22,5°C	85 %	0.35 mm	Yes
6	19,8°C	30 %	0.05 mm	No
7	20,2°C	80 %	0.31 mm	Yes

Tabla 5: Weather Data for kNN Classification

Ejercicio 23 (Feature Types (2 pts.))**.** Which features in Table 5 are categorical and which are nominal?

Ejercicio 24 (Distance Metrics (5 pts.))**.** Give two different distance metrics and provide their mathematical formulas. Explain the formulas and the differences between them.

Ejercicio 25 (kNN Execution (8 pts.))**.** Classify the two days given in Table 6 using the kNN algorithm with $k = 4$.

Day	**Max. Temperature**	**Cloudiness**	**Rain Level**
1	21,3°C	45 %	0.25 mm
2	18,3°C	25 %	0.00 mm

Tabla 6: Data to Classify

Hint: Use the following modified distance metric:

$$d(\text{Day 1}, \text{Day 2}) = |\text{Difference in Cloudiness (\%)}| + 100 \cdot |\text{Difference in Rain Level (mm)}|$$

Example:

$$d(\text{Day 1}, \text{Day 2}) = |30 - 80| + 100 \cdot |0{,}30 - 0{,}28| = 50 + 100 \cdot 0{,}02 = 50 + 2 = 52$$

Specify the distance ranking explicitly for both days and state the final classification decision.

Part 3: Decision Tree & Random Forest (20 pts.)

Ejercicio 26 (ID3 Algorithm Steps (4 pts.)). Name and explain the steps of the ID3 algorithm.

Ejercicio 27 (ID3 vs. C4.5 (2 pts.)). Name one difference between the ID3 and C4.5 algorithms.

Ejercicio 28 (Gini Impurity Calculation (3 pts.)). Consider the dataset with 20 points given in Table 7 and illustrated in Figure 3. Features F_1 and F_2 correspond to $Feature1$ and $Feature2$. Calculate the individual Gini impurities and the total Gini impurity for the split at $F_2 = 4$ based on the target class *Type* (Pear/Apple).

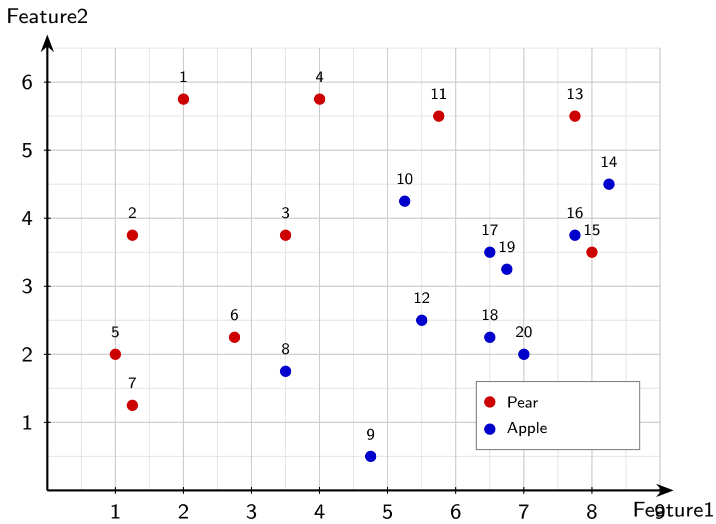

Figura 3: Scatter Plot of Feature1 vs Feature2

Ejercicio 29 (Decision Tree Construction (5 pts.)). Table 8 provides several candidate splits and their corresponding Gini Impurities. Create a Decision Tree that correctly classifies all points based on this table. Draw it explicitly and specify its depth.

Ejercicio 30 (Bootstrap Samples (1 pt.)). Why are bootstrap samples used when building a Random Forest?

Ejercicio 31 (Random Forest vs. Decision Tree Classification (5 pts.)). Classify the point $(F_1 = 4, F_2 = 5{,}25)$ using the Random Forest in Figure 4. State the prediction of each individual tree as well as the aggregated output. Then, classify the same point using the large Decision Tree on the right (draw the decision path).

Index	F_1	F_2	**Type**
1	2.00	5.75	Pear
2	1.25	3.75	Pear
3	3.50	3.75	Pear
4	4.00	5.75	Pear
5	1.00	2.00	Pear
6	2.75	2.25	Pear
7	1.25	1.25	Pear
8	3.50	1.75	Apple
9	4.75	0.50	Apple
10	5.25	4.25	Apple
11	5.75	5.50	Pear
12	5.50	2.50	Apple
13	7.75	5.50	Pear
14	8.25	4.50	Apple
15	8.00	3.50	Pear
16	7.75	3.75	Apple
17	6.50	3.50	Apple
18	6.50	2.25	Apple
19	6.75	3.25	Apple
20	7.00	2.00	Apple

Tabla 7: Fruit Dataset

Feature	**Split**	**Gini Impurity**
F_1	< 1	0.487
F_1	≤ 2	0.469
F_1	< 3	0.313
F_1	< 4	0.444
F_1	< 5	0.374
F_1	< 6	0.487
F_1	≤ 7	0.500
F_1	< 8	0.421
F_2	< 1	0.498
F_2	< 2	0.493
F_2	< 3	0.479
F_2	< 4	0.493
F_2	< 5	0.375

Tabla 8: Feature Splits with Gini Impurity

Part 4: Encoding & Forward Propagation (5 pts.)

Ejercicio 32 (Ordinal Encoding (2 pts.)). Given a table column with the elements {“OH”, “OR”, “FI”, “NY”, “WY”, “WI”}, perform ordinal encoding on these expressions and present the output in a table. What happens if you now want to add “CA” to the list?

Ejercicio 33 (Forward Propagation Network (2 pts.)). Given the small neural net-

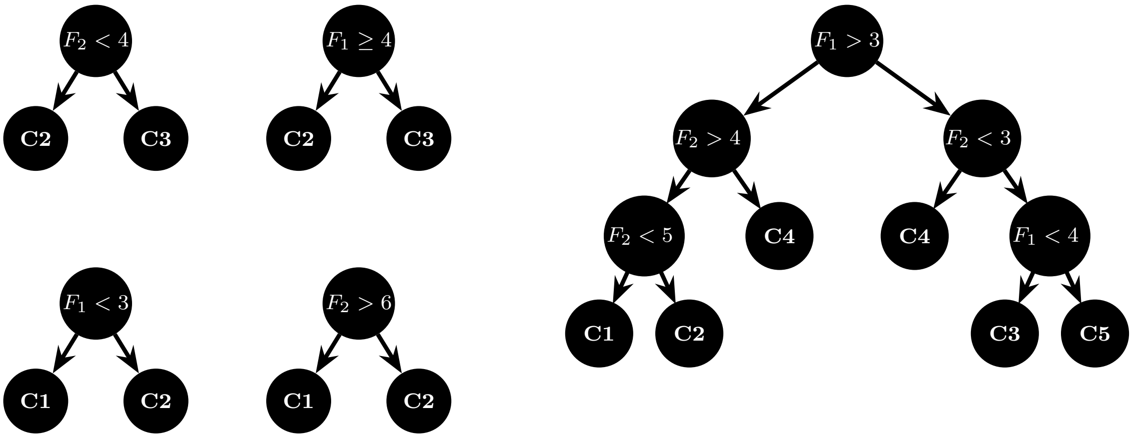

Figura 4: Random Forest & Decision Tree

work below with input $x = 12$, calculate the output values after applying the activation functions and write them into the corresponding white boxes.

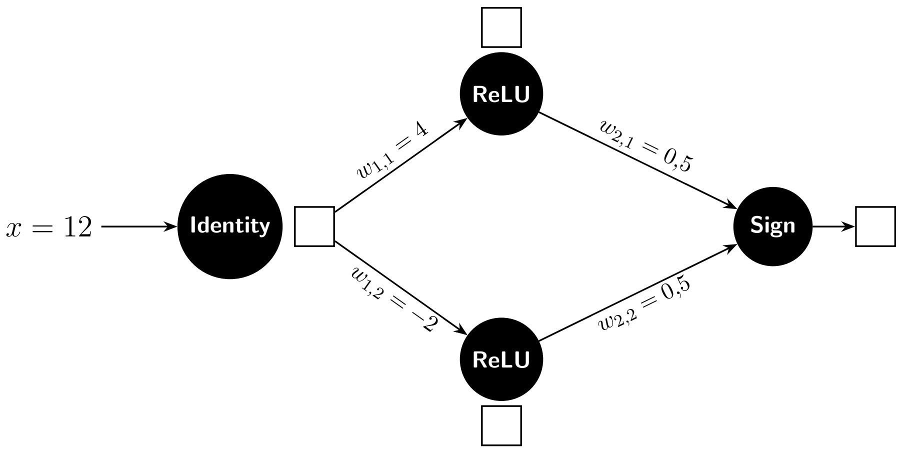

Figura 5: Small Feedforward Neural Network

Ejercicio 34 (Error Calculation - MAE & MSE (1 pt.))**.** Suppose the neural network outputs 2, and the target value should be 4. Calculate the Mean Absolute Error (MAE) and Mean Squared Error (MSE).

Part 1: Short Questions (54 pts.)

Ejercicio 1 (Unsupervised Learning (3 pts.)). What is unsupervised learning? Mark whether each of the following methods belongs to unsupervised learning or not.

Method	Right	Wrong
PCA	☒	☐
Hill Climbing	☐	☒
kNN	☐	☒
kMeans	☒	☐
t-SNE	☒	☐
GANs	☒	☐

Tabla 9: Classification of Unsupervised Learning Methods

Ejercicio 2 (KDD Process (2 pts.)). What are the steps involved in the Knowledge Discovery in Databases (KDD) process?

Ejercicio 3 (Date Conversion Issues (1 pt.)). Indicate a possible problem when converting the date 08/01/23.

It can either be understood as August 1, 2023, or January 8, 2023, depending on the date format used (MM/DD/YY vs. DD/MM/YY).

Ejercicio 4 (Basic Statistics (3 pts.)). Given the numbers $A = \{3, 4, 5, 10, 11, 1, 8\}$, compute the median, mean, and variance.

- For the median, sort the numbers: $1, 3, 4, 5, 8, 10, 11$. The median is the middle value: 5.
- For the mean, calculate the sum of the numbers and divide by the count:

$$\text{Mean} = \frac{3 + 4 + 5 + 10 + 11 + 1 + 8}{7} = \frac{42}{7} = 6.$$

- For the variance, the formula is:

$$\begin{aligned}\text{Variance} &= \frac{1}{n}\sum_{i=1}^{n}(x_i - \text{Mean})^2 \\ &= \frac{1}{7}\left[3^2 + 2^2 + 1^2 + 4^2 + 5^2 + 5^2 + 2^2\right] \\ &= \frac{84}{7} = 12\end{aligned}$$

Ejercicio 5 (Data Aggregation Problems (2 pts.)). What problems can occur when performing data aggregation? Mark whether the statement is right or wrong.

Ejercicio 6 (Curse of Dimensionality (2 pts.)). Where does the effect of the "Curse of Dimensionality" occur?

Problem	Right	Wrong
Information gain	☐	☒
Information loss	☒	☐
Data reduction	☒	☐
Data separation	☐	☒

Tabla 10: Data Aggregation Issues

Method	Right	Wrong
High-dimensional search	☒	☐
Data aggregation	☐	☒
Optimize a neural network	☒	☐
Execute a binary classifier	☐	☒

Tabla 11: Curse of Dimensionality Contexts

Ejercicio 7 (Search Methods (1 pt.)). For which search method is Depth-Limited Search a modification?

Depth-Limited Search is a modification of Depth-First Search (DFS), where a depth limit is imposed to prevent the search from going too deep into the search tree, which can help avoid infinite loops in cases of cyclic graphs.

Ejercicio 8 (Topological Search (5 pts.)). Give an example to explain Topological Search.

Topological Search is used in directed acyclic graphs (DAGs) to order the vertices such that for every directed edge from vertex A to vertex B, A comes before B in the ordering. An example could be deciding the order in which a student has to take courses based on prerequisites. If Course A is a prerequisite for Course B, then in the topological order, Course A will appear before Course B.

Ejercicio 9 (Genetic Algorithms - Mutation & Crossover (3 pts.)). Given two genes of length 10 with bases $\{0, 1, 2\}$:

- Gene 1: 0101211221
- Gene 2: 2111111001

Apply mutation and a crossover type to these genes starting with Gene 1.

Parent 1	0	1	0	1	2	1	1	2	2	1
Parent 2	2	1	1	1	1	1	1	0	0	1
Offspring	2	1	0	1	2	1	1	0	0	1
Swap Mutation	2	1	0	1	2	1	1	0	**1**	**0**

Tabla 12: Example of Order-Based Crossover (OX2), where the fixed positions are 1, 2, 4, and 6.

Ejercicio 10 (Uniform Cost Search (4 pts.)). Explain Uniform Cost Search using the graph in Figure 6. Go through the different steps.

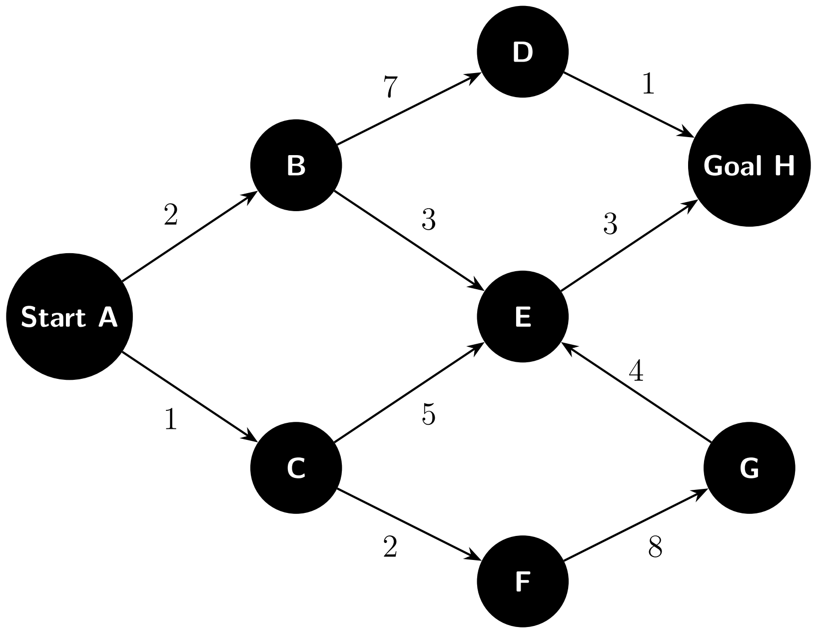

Figura 6: Uniform Cost Graph

A Uniform Cost Search (UCS) is an algorithm used to find the least-cost path from a starting node to a goal node in a weighted graph. It expands the node with the lowest cumulative cost first, ensuring that the path found is optimal.

- Start at node A with a cumulative cost of 0. The priority queue contains: $[(A, 0)]$.
- Expand node A. The neighbors are B (cost 2) and C (cost 1). Add them to the queue: $[(C, 1), (B, 2)]$.
- Expand node C (lowest cost 1). The neighbors are E (cost 6) and F (cost 3). Add them to the queue: $[(B, 2), (F, 3), (E, 6)]$.
- Expand node B (lowest cost 2). The neighbors are D (cost 9) and E (cost 5). Add them to the queue: $[(F, 3), (E, 5), (D, 9)]$.
- Expand node F (lowest cost 3). The neighbor is G (cost 11). Add it to the queue: $[(E, 5), (D, 9), (G, 11)]$.
- Expand node E (lowest cost 5). The neighbors are H (cost 8). Add it to the queue: $[(H, 8), (D, 9), (G, 11)]$.
- Expand node H (lowest cost 8). Since H is the goal node, the search terminates, and the least-cost path from A to H is found with a total cost of 8.

Ejercicio 11 (Roulette Selection (2 pts.))**.** What is Roulette Selection in genetic algorithms?

Roulette Selection is a selection method used in genetic algorithms to choose individuals from a population for reproduction based on their fitness. Each individual is assigned a probability of being selected that is proportional to its fitness score. The higher the fitness, the greater the chance of being selected. This is often visualized as a roulette wheel where each individual's segment size corresponds to its fitness, and a random spin determines which individual is chosen for the next generation.

This has however one big disadvantage: if one individual has a very high fitness compared to the others, it can dominate the selection process, leading to premature convergence and loss of genetic diversity in the population. Rank selection is a better alternative in such cases, as it assigns selection probabilities based on the rank of individuals rather than their absolute fitness values, ensuring a more balanced selection process.

Ejercicio 12 (Underfitting vs. Overfitting (2 pts.))**.** Explain what is meant by underfitting and overfitting in machine learning.

Underfitting occurs when a machine learning model is too simple to capture the underlying patterns in the training data, resulting in poor performance on both the training and test datasets. It fails to learn the relationships between features and target variables, leading to high bias and low variance.

Overfitting, on the other hand, happens when a model is too complex and learns not only the underlying patterns but also the noise in the training data. This results in excellent performance on the training dataset but poor generalization to unseen test data. Overfitting is characterized by low bias and high variance, where the model captures random fluctuations rather than the true signal.

Ejercicio 13 (Confusion Matrix Elements (6 pts.))**.** Fill in the missing entries correctly in Table 13 and explain the elements of the confusion matrix.

Predicted / Actual	**Actual Yes**	**Actual No**	**Actual Total**
Predicted Yes	45 %	15 %	60 %
Predicted No	25 %	15 %	40 %
Predicted Total	70 %	30 %	100 %

Tabla 13: Confusion Matrix

- Predicted Yes / Actual Yes (True Positives): 45 % - The model correctly predicted positive cases.
- Predicted Yes / Actual No (False Positives): 15 % - The model incorrectly predicted positive cases when they were actually negative.
- Predicted No / Actual Yes (False Negatives): 25 % - The model incorrectly predicted negative cases when they were actually positive.
- Predicted No / Actual No (True Negatives): 15 % - The model correctly predicted negative cases.

Ejercicio 14 (F1-Score Definition (1 pt.))**.** How is the F1-Score defined?

It is the harmonic mean of precision and recall, calculated as:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Ejercicio 15 (Resampling Methods (2 pts.))**.** Name one resampling method and explain how it works.

One resampling method is k-fold cross-validation. In k-fold cross-validation, the dataset is divided into k equal-sized subsets or “folds”. The model is trained k times, each time using k-1 folds for training and the remaining fold for validation. This process is repeated until each fold has been used as the validation set once. The performance metrics are then averaged over the k iterations to provide a more robust estimate of the model’s generalization ability.

Ejercicio 16 (Linear vs. Ridge vs. Lasso (3 pts.))**.** List the differences between Linear Regression, Ridge Regression, and Lasso Regression.

The only difference lies on the loss function used to train the model.

- Linear Regression: Uses the Ordinary Least Squares (OLS) loss function, which minimizes the sum of squared residuals between the predicted and actual values.
$$\text{Loss}_{\text{OLS}} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$
- Ridge Regression: Adds an L2 regularization term to the OLS loss function, which penalizes large coefficients to prevent overfitting. The loss function is:
$$\text{Loss}_{\text{Ridge}} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$
- Lasso Regression: Adds an L1 regularization term to the OLS loss function, which can shrink some coefficients to zero, effectively performing feature selection. The loss function is:
$$\text{Loss}_{\text{Lasso}} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

Ejercicio 17 (Principal Component Analysis (1 pt.))**.** What happens during Principal Component Analysis (PCA)?

During Principal Component Analysis (PCA), the original dataset is transformed into a new set of orthogonal axes called principal components. These components are linear combinations of the original features and are ordered by the amount of variance they capture from the data. The first principal component captures the maximum variance, followed by the second, and so on. PCA reduces the dimensionality of the data while retaining as much variability as possible, which can help in visualization, noise reduction, and improving model performance.

Ejercicio 18 (Top-Down Approach (2 pts.))**.** What is the top-down approach in hierarchical clustering or decision trees?

The top-down approach, also known as divisive clustering, starts with all data points in a single cluster and recursively splits them into smaller clusters.

In decision trees, the top-down approach begins with the entire dataset at the root node and recursively partitions it into subsets based on feature values, creating branches and nodes until a stopping criterion is met (e.g., maximum depth, minimum samples per leaf, or pure nodes). This method allows for a hierarchical representation of the data, where each split aims to maximize the separation of classes or minimize impurity.

Ejercicio 19 (Linkage Types in Hierarchical Clustering (4 pts.)). Name two types of linkage in hierarchical clustering and state the mutual advantages and disadvantages of each.

- Single Linkage: This method defines the distance between two clusters as the minimum distance between any single pair of points in the two clusters.

  $$d_{\text{single}}(C_1, C_2) = \min_{x \in C_1, y \in C_2} d(x, y)$$

  It is sensitive to noise and outliers, which can lead to the "chaining effect," where clusters may be merged based on a single close point, resulting in elongated and less compact clusters. However, it can capture non-elliptical shapes and is computationally efficient.

- Complete Linkage: This method defines the distance between two clusters as the maximum distance between any single pair of points in the two clusters.

  $$d_{\text{complete}}(C_1, C_2) = \max_{x \in C_1, y \in C_2} d(x, y)$$

  It tends to produce more compact and spherical clusters, making it less sensitive to noise and outliers compared to single linkage. However, it may fail to capture elongated or irregularly shaped clusters and can be computationally more intensive.

Ejercicio 20 (Activation Function Definition (1 pt.)). Name an activation function and give its mathematical definition.

The Rectified Linear Unit (ReLU) is a widely used activation function in neural networks. It is defined mathematically as:

$$\text{ReLU}(x) = \max(0, x)$$

This function outputs the input directly if it is positive; otherwise, it outputs zero. ReLU introduces non-linearity into the model, allowing it to learn complex patterns, and helps mitigate the vanishing gradient problem during training.

Ejercicio 21 (Feedback Connections in Neural Networks (3 pts.)). Draw and label a lateral feedback connection, a direct feedback connection, and an indirect feedback connection in the network given in Figure 7.

Ejercicio 22 (Stemming (1 pt.)). What is stemming in Natural Language Processing (NLP)?

Stemming is a text normalization technique in Natural Language Processing (NLP) that reduces words to their base or root form, known as the stem. The process involves removing suffixes and prefixes from words to group together different forms

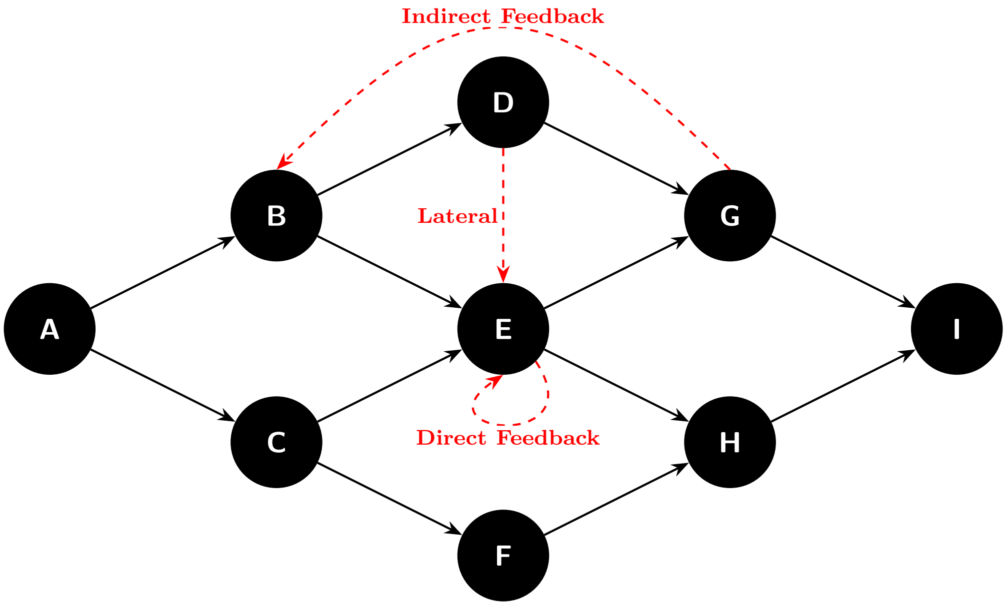

Figura 7: Forward-Feeding Network

of the same word, which helps in reducing the dimensionality of the text data and improving the performance of NLP models. For example, the words "running," or "runner," can be reduced to the stem "run." Opposite to lemmatization, stemming does not guarantee that the resulting stem is a valid word in the language; it may produce non-dictionary forms.

Part 2: kNN Classification (15 pts.)

The precipitation dataset (Table 14) is provided. The goal is to classify the two days from Table 15 as rainy or non-rainy days.

Day	Max. Temperature	Cloudiness	Rain Level	Rainy Day
1	20,1°C	30 %	0.30 mm	Yes
2	29,8°C	80 %	0.28 mm	Yes
3	21,2°C	25 %	0.00 mm	No
4	25,0°C	20 %	0.01 mm	No
5	22,5°C	85 %	0.35 mm	Yes
6	19,8°C	30 %	0.05 mm	No
7	20,2°C	80 %	0.31 mm	Yes

Tabla 14: Weather Data for kNN Classification

Ejercicio 23 (Feature Types (2 pts.))**.** Which features in Table 14 are categorical and which are nominal?

Among all the features in Table 14, the **Rainy Day** feature is categorical and nominal, as it represents a binary classification (Yes/No) without any inherent order. The other features (**Max. Temperature**, **Cloudiness**, and **Rain Level**) are numerical and continuous, representing measurable quantities.

Ejercicio 24 (Distance Metrics (5 pts.))**.** Give two different distance metrics and provide their mathematical formulas. Explain the formulas and the differences between them.

Two commonly used distance metrics are the Euclidean distance and the Manhattan distance.

- **Euclidean Distance:** This metric calculates the straight-line distance between two points in a multi-dimensional space. The formula for Euclidean distance between two points $P = (p_1, p_2, ..., p_n)$ and $Q = (q_1, q_2, ..., q_n)$ is:

$$d_{\text{Euclidean}}(P, Q) = \sqrt{\sum_{i=1}^{n} (p_i - q_i)^2}$$

  It is sensitive to the scale of the features and can be influenced by outliers.

- **Manhattan Distance:** Also known as the L1 distance or taxicab distance, this metric calculates the sum of the absolute differences between the coordinates of two points. The formula for Manhattan distance is:

$$d_{\text{Manhattan}}(P, Q) = \sum_{i=1}^{n} |p_i - q_i|$$

  It is less sensitive to outliers compared to Euclidean distance and is often used when the data has a grid-like structure.

The main difference between the two metrics is that Euclidean distance measures the shortest path (straight line) between points, while Manhattan distance measures the total distance traveled along axes at right angles (like navigating a grid). This can lead to different distance calculations and neighbor selections in kNN, especially in high-dimensional spaces or when features have different scales.

Ejercicio 25 (kNN Execution (8 pts.)). Classify the two days given in Table 15 using the kNN algorithm with $k = 4$.

Day	**Max. Temperature**	**Cloudiness**	**Rain Level**
1	21,3°C	45 %	0.25 mm
2	18,3°C	25 %	0.00 mm

Tabla 15: Data to Classify

Hint: Use the following modified distance metric:

$$d(\text{Day 1}, \text{Day 2}) = |\text{Difference in Cloudiness (\%)}| + 100 \cdot |\text{Difference in Rain Level (mm)}|$$

Example:

$$d(\text{Day 1}, \text{Day 2}) = |30 - 80| + 100 \cdot |0{,}30 - 0{,}28| = 50 + 100 \cdot 0{,}02 = 50 + 2 = 52$$

Specify the distance ranking explicitly for both days and state the final classification decision.

The distance calculations for Day 1 and Day 2 in Table 15 against the dataset in Table 14 are as follows:

- For Day 1 (21,3°C, 45 %, 0,25 mm):
    - Distance to Day 1: $|30 - 45| + 100 \cdot |0{,}30 - 0{,}25| = 15 + 5 = 20$
    - Distance to Day 2: $|80 - 45| + 100 \cdot |0{,}28 - 0{,}25| = 35 + 3 = 38$
    - Distance to Day 3: $|25 - 45| + 100 \cdot |0{,}00 - 0{,}25| = 20 + 25 = 45$
    - Distance to Day 4: $|20 - 45| + 100 \cdot |0{,}01 - 0{,}25| = 25 + 24 = 49$
    - Distance to Day 5: $|85 - 45| + 100 \cdot |0{,}35 - 0{,}25| = 40 + 10 = 50$
    - Distance to Day 6: $|30 - 45| + 100 \cdot |0{,}05 - 0{,}25| = 15 + 20 = 35$
    - Distance to Day 7: $|80 - 45| + 100 \cdot |0{,}31 - 0{,}25| = 35 + 6 = 41$

  Therefore, the sorted distances for Day 1 are shown in Table 16. Since only the first $k = 4$ rows are considered, the majority of the nearest neighbors are rainy days (3 out of 4), so Day 1 is classified as a rainy day.

- For Day 2 (18,3°C, 25 %, 0,00 mm):
    - Distance to Day1: $|30 - 25| + 100 \cdot |0{,}30 - 0{,}00| = 5 + 30 = 35$
    - Distance to Day2: $|80 - 25| + 100 \cdot |0{,}28 - 0{,}00| = 55 + 28 = 83$
    - Distance to Day3: $|25 - 25| + 100 \cdot |0{,}00 - 0{,}00| = 0 + 0 = 0$

Day	Distance	Rainy Day
1	20	Yes
6	35	No
2	38	Yes
7	41	Yes
3	45	No
4	49	No
5	50	Yes

Tabla 16: Distance Ranking for Day 1

- Distance to Day4: $|20 - 25| + 100 \cdot |0{,}01 - 0{,}00| = 5 + 1 = 6$
- Distance to Day5: $|85 - 25| + 100 \cdot |0{,}35 - 0{,}00| = 60 + 35 = 95$
- Distance to Day6: $|30 - 25| + 100 \cdot |0{,}05 - 0{,}00| = 5 + 5 = 10$
- Distance to Day7: $|80 - 25| + 100 \cdot |0{,}31 - 0{,}00| = 55 + 31 = 86$

Therefore, the sorted distances for Day 2 are shown in Table 17. Since only

Day	Distance	Rainy Day
3	0	No
4	6	No
6	10	No
1	35	Yes
2	83	Yes
7	86	Yes
5	95	Yes

Tabla 17: Distance Ranking for Day 2

the first $k = 4$ rows are considered, the majority of the nearest neighbors are non-rainy days (3 out of 4), so Day 2 is classified as a non-rainy day.

Part 3: Decision Tree & Random Forest (20 pts.)

Ejercicio 26 (ID3 Algorithm Steps (4 pts.))**.** Name and explain the steps of the ID3 algorithm.

The ID3 (Iterative Dichotomiser 3) algorithm is a decision tree learning algorithm that constructs a decision tree by employing a top-down, greedy approach. The steps of the ID3 algorithm are as follows:

1. **Select the Best Attribute:** For each node, for each attribute, calculate the entropy of the dataset resulting from splitting the data based on that attribute. The attribute with the lowest entropy is selected as the best attribute to split the data at that node.

2. **Create a Decision Node:** Create a decision node in the tree that corresponds to the selected attribute. This node will have branches for each possible value of the attribute.

3. **Split the Dataset:** Partition the dataset into subsets based on the values of the selected attribute. Each subset corresponds to a branch of the decision node.

4. **Repeat Recursively:** For each subset, repeat the process of selecting the best attribute and creating decision nodes until one of the following stopping criteria is met:

   - All instances in the subset belong to the same class (pure node).
   - There are no remaining attributes to split on.
   - A predefined stopping condition is reached (e.g., maximum depth, minimum samples per leaf).

5. **Assign Class Labels:** Once a stopping criterion is met, assign the majority class label of the instances in the subset to the leaf node.

Ejercicio 27 (ID3 vs. C4.5 (2 pts.))**.** Name one difference between the ID3 and C4.5 algorithms.

One main difference between the ID3 and C4.5 algorithms is that C4.5 can handle both continuous and categorical attributes, while ID3 can only handle categorical attributes. C4.5 uses a method called “thresholding” to determine the best split point for continuous attributes, allowing it to create decision trees that can work with a wider variety of data types.

Ejercicio 28 (Gini Impurity Calculation (3 pts.))**.** Consider the dataset with 20 points given in Table 18 and illustrated in Figure 8. Features F_1 and F_2 correspond to $Feature1$ and $Feature2$. Calculate the individual Gini impurities and the total Gini impurity for the split at $F_2 = 4$ based on the target class *Type* (Pear/Apple).

When splitting the dataset at $F_2 = 4$, we have two subsets:

- **Subset 1 ($F_2 < 4$):** The dataset here contains 6 pears (indices 2, 3, 5, 6, 7, 15) and 8 apples (indices 8, 9, 12, 16, 17, 18, 19, 20). The Gini impurity for this subset is calculated as:

$$Gini_1 = 1 - \left(\frac{6}{14}\right)^2 - \left(\frac{8}{14}\right)^2 = \frac{24}{49} \approx 0{,}4898$$

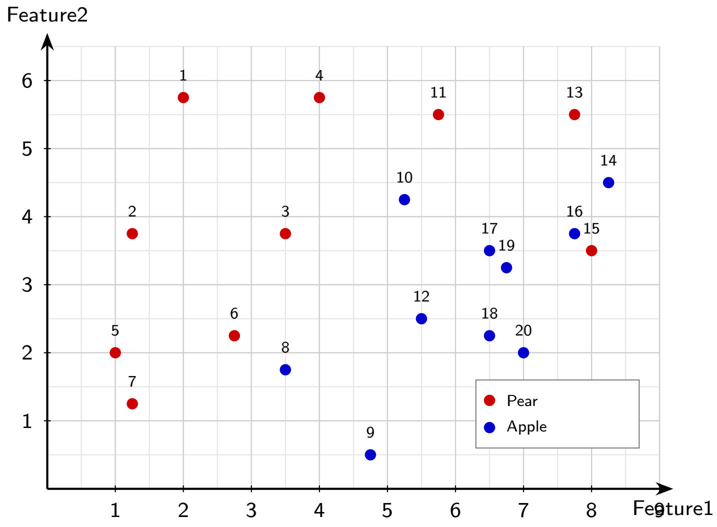

Figura 8: Scatter Plot of Feature1 vs Feature2

Index	F_1	F_2	**Type**
1	2.00	5.75	Pear
2	1.25	3.75	Pear
3	3.50	3.75	Pear
4	4.00	5.75	Pear
5	1.00	2.00	Pear
6	2.75	2.25	Pear
7	1.25	1.25	Pear
8	3.50	1.75	Apple
9	4.75	0.50	Apple
10	5.25	4.25	Apple
11	5.75	5.50	Pear
12	5.50	2.50	Apple
13	7.75	5.50	Pear
14	8.25	4.50	Apple
15	8.00	3.50	Pear
16	7.75	3.75	Apple
17	6.50	3.50	Apple
18	6.50	2.25	Apple
19	6.75	3.25	Apple
20	7.00	2.00	Apple

Tabla 18: Fruit Dataset

- **Subset 2 ($F_2 \geqslant 4$):** The dataset here contains 4 pears (indices 1, 4, 11, 13) and 2 apples (indices 10, 14). The Gini impurity for this subset is calculated as:

$$Gini_2 = 1 - \left(\frac{4}{6}\right)^2 - \left(\frac{2}{6}\right)^2 = \frac{4}{9} \approx 0{,}4444$$

The total Gini impurity for the split at $F_2 = 4$ is calculated as a weighted average of the individual Gini impurities:

$$Gini_{total} = \frac{14}{20} \cdot Gini_1 + \frac{6}{20} \cdot Gini_2 = \frac{14}{20} \cdot \frac{24}{49} + \frac{6}{20} \cdot \frac{4}{9} = \frac{10}{21} \approx 0{,}4762$$

Ejercicio 29 (Decision Tree Construction (5 pts.))**.** Table 19 provides several candidate splits and their corresponding Gini Impurities. Create a Decision Tree that correctly classifies all points based on this table. Draw it explicitly and specify its depth.

Feature	**Split**	**Gini Impurity**
F_1	< 1	0.487
F_1	≤ 2	0.469
F_1	< 3	0.313
F_1	< 4	0.444
F_1	< 5	0.374
F_1	< 6	0.487
F_1	≤ 7	0.500
F_1	< 8	0.421
F_2	< 1	0.498
F_2	< 2	0.493
F_2	< 3	0.479
F_2	< 4	0.493
F_2	< 5	0.375

Tabla 19: Feature Splits with Gini Impurity

Ejercicio 30 (Bootstrap Samples (1 pt.))**.** Why are bootstrap samples used when building a Random Forest?

Bootstrap samples are used in Random Forest to introduce diversity among the individual decision trees in the ensemble. By randomly sampling the training data with replacement, each tree is trained on a different subset of the data, which helps to reduce overfitting and improve generalization. This randomness allows the Random Forest to capture a wider variety of patterns in the data, leading to more robust and accurate predictions when aggregating the outputs of all trees.

Ejercicio 31 (Random Forest vs. Decision Tree Classification (5 pts.))**.** Classify the point $(F_1 = 4, F_2 = 5{,}25)$ using the Random Forest in Figure 9. State the prediction of each individual tree as well as the aggregated output. Then, classify the same point using the large Decision Tree on the right (draw the decision path).

In the random forest, there are four trees. The predictions for the point $(F_1 = 4, F_2 = 5{,}25)$ are as follows:

- Tree 1: C3
- Tree 2: C2
- Tree 3: C2
- Tree 4: C2

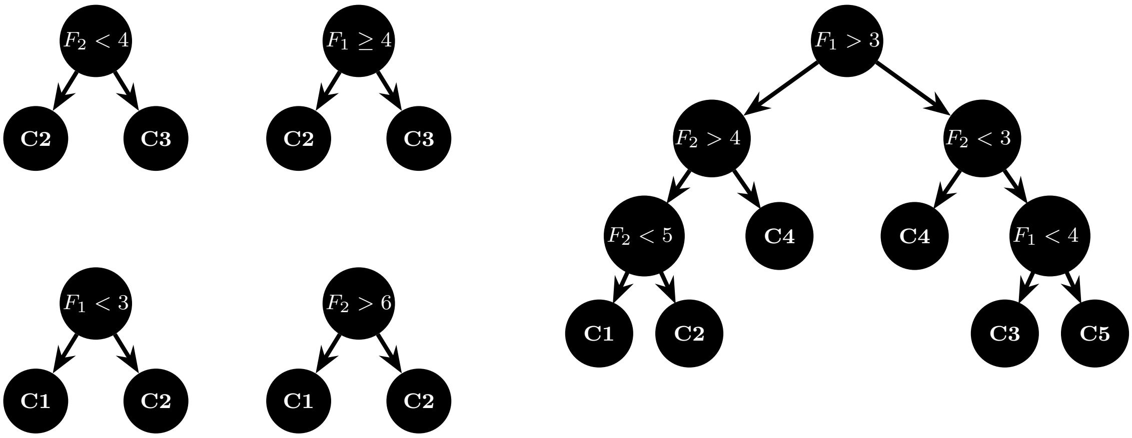

Figura 9: Random Forest & Decision Tree

The aggregated output is C2, as it is the most frequent prediction among the four trees.

In the large decision tree, the decision path for the point $(F_1 = 4, F_2 = 5{,}25)$ is as follows:

1. Start at the root node: $F_1 > 3$ (True)
2. Move to the left child: $F_2 > 4$ (True)
3. Move to the left child: $F_2 < 5$ (False)
4. Move to the right child: C2

Therefore, the classification for the point $(F_1 = 4, F_2 = 5{,}25)$ using the large decision tree is also C2.

Part 4: Encoding & Forward Propagation (5 pts.)

Ejercicio 32 (Ordinal Encoding (2 pts.)). Given a table column with the elements {"OH", "OR", "FI", "NY", "WY", "WI"}, perform ordinal encoding on these expressions and present the output in a table. What happens if you now want to add "CA" to the list?

The ordinal encoding for the given elements can be represented as shown in Table 20.

Element	Ordinal Encoding
FI	1
OH	2
OR	3
NY	4
WI	5
WY	6

Tabla 20: Ordinal Encoding of Elements

If we want to add "CA" to the list, it would have to be assigned the first ordinal value (1), and all other elements would need to be shifted accordingly. This would change the encoding of all existing elements, which can lead to inconsistencies and misinterpretations in the data representation.

Ejercicio 33 (Forward Propagation Network (2 pts.)). Given the small neural network below with input $x = 12$, calculate the output values after applying the activation functions and write them into the corresponding white boxes.

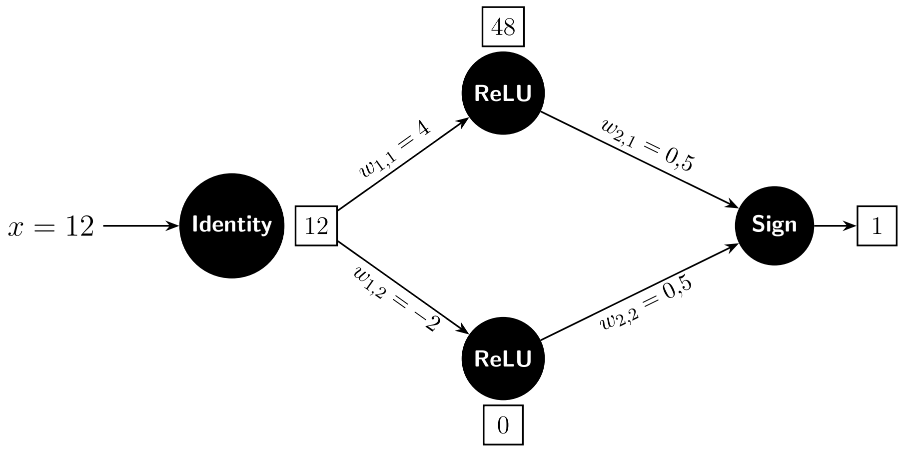

Figura 10: Small Feedforward Neural Network

Ejercicio 34 (Error Calculation - MAE & MSE (1 pt.)). Suppose the neural network outputs 2, and the target value should be 4. Calculate the Mean Absolute Error (MAE) and Mean Squared Error (MSE).

The Mean Absolute Error (MAE) is calculated as the absolute difference between the predicted value and the target value:

$$\text{MAE} = |2 - 4| = 2$$

The Mean Squared Error (MSE) is calculated as the square of the difference between the predicted value and the target value:

$$\text{MSE} = (2 - 4)^2 = (-2)^2 = 4$$